Recap

① w/o critical cases i/p → (LTI) → o/p ← wire/curve on any branch

single signal

Imp. res. $h(t) \Leftrightarrow H(s)$ ← Laplace

② BIBO Stsky if all poles (s₀ when $H(s_0)=\infty$) strictly in left half of complex plane.

↳ all passive circuits w/ "resistance" are stsky

w/o res. LC can be unstsky

poles on Im axis

(PS1)

(PS2)

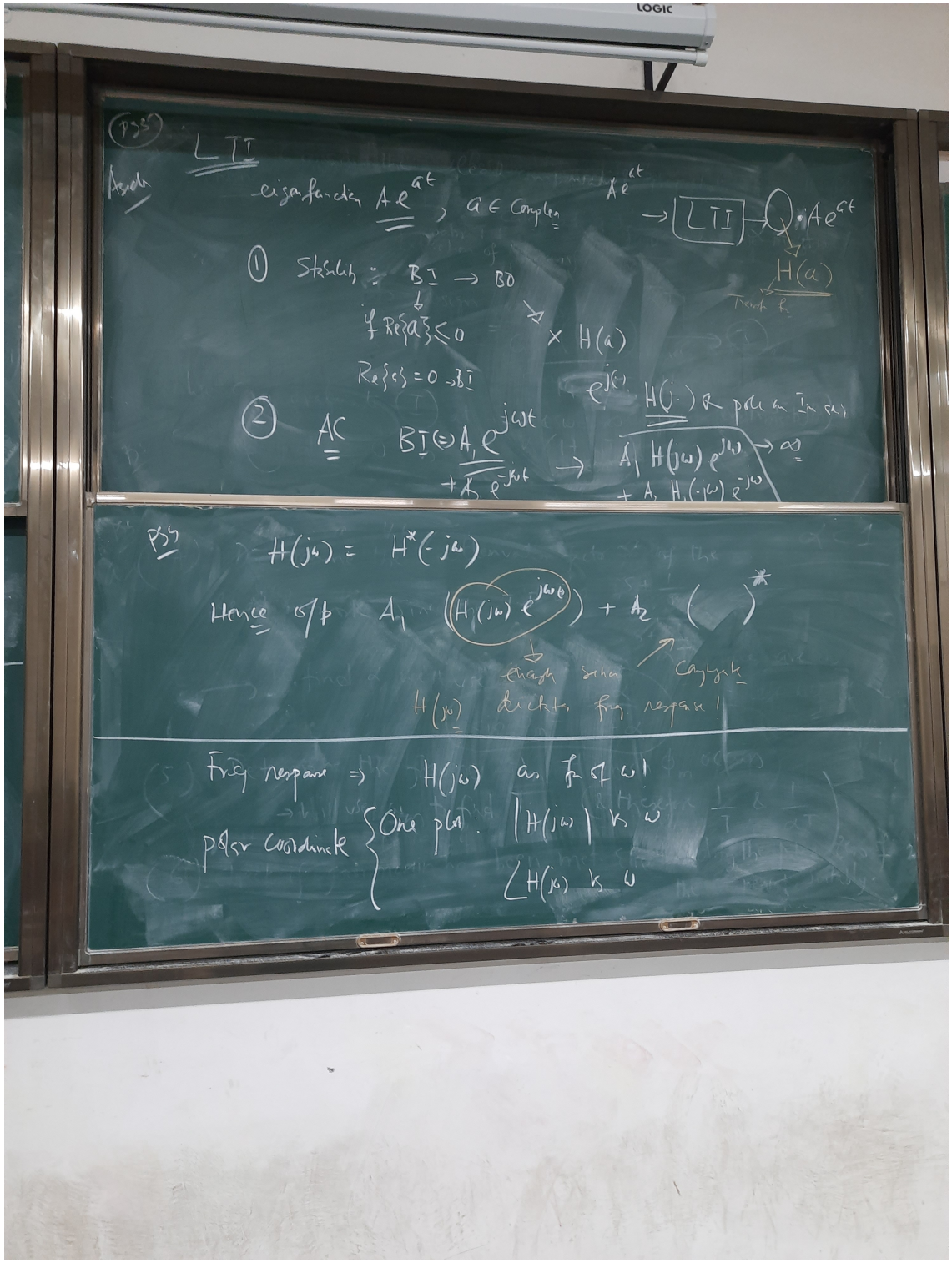
③ Recall AC steady state response = AC

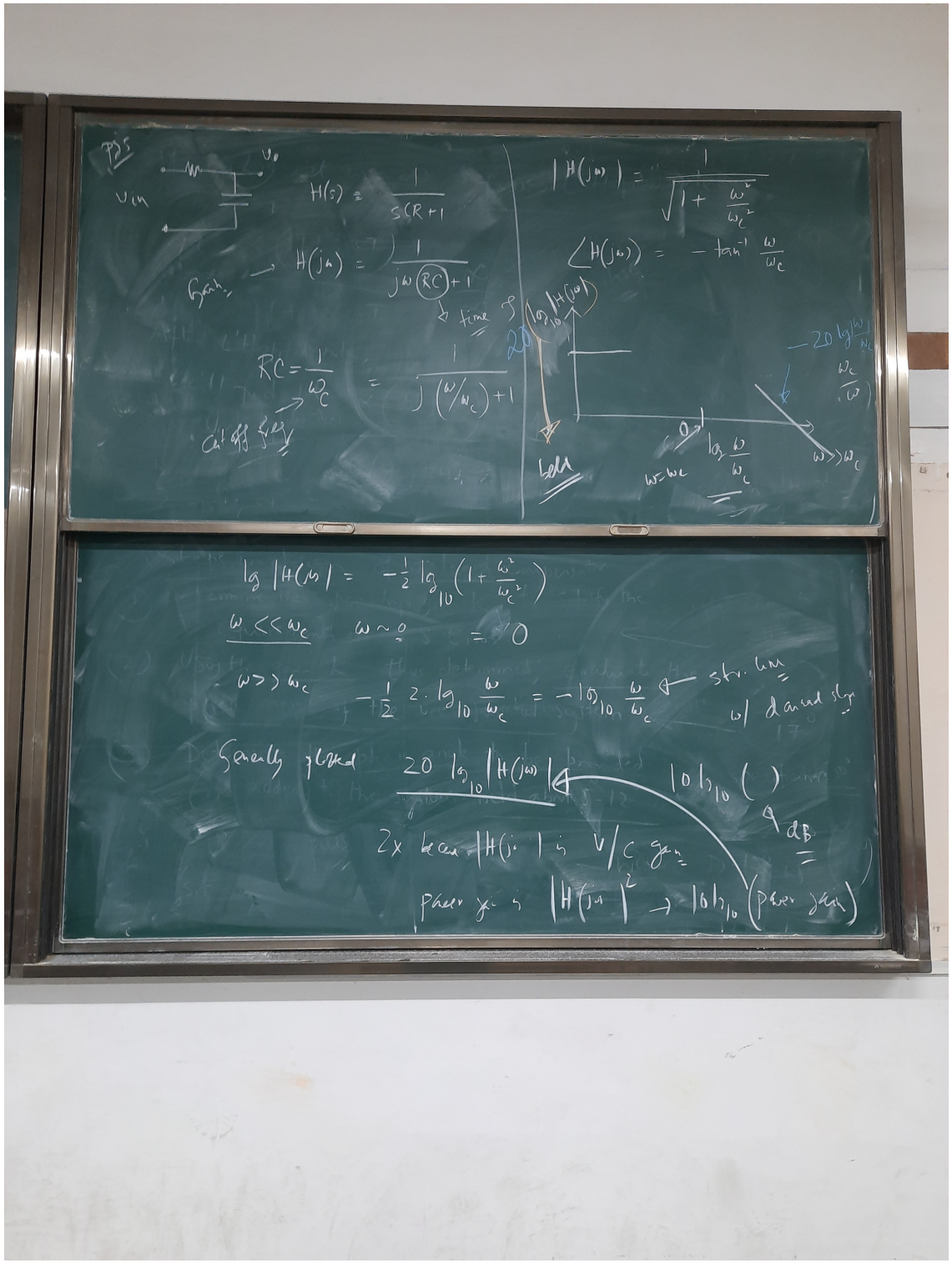
$H(s)\big|_{s=j\omega}$ gives steady state response

For ② & ③ a few proofs/Eg. we discussed!

→ For ③, there is a simple computational alg!

→ For ②, ... partial fr. alg!


LTI
eigenfunction Ae^{at}, $a \in$ Complex
$Ae^{at} \rightarrow$ LTI $\rightarrow Q \cdot Ae^{at}$
$H(a)$
Transfer fn
① Stability: BI $\rightarrow$ BO
if $Re\{a\} \leq 0$
$Re\{a\} = 0$ → BI
$\times H(a)$
$e^{j\omega t}$
② AC
$BI = A_1 e^{j\omega t} + A_2 e^{-j\omega t}$
$H(j\cdot)$ R part as Im sin
$A_1 H(j\omega) e^{j\omega t} + A_2 H(-j\omega) e^{-j\omega t}$
$\rightarrow \infty$
PSD
$H(j\omega) = H^*(-j\omega)$
Hence o/p $A_1 (H(j\omega) e^{j\omega t}) + A_2 (\quad)^*$
Enough when Conjugate
$H(j\omega)$ decides freq response
Freq response $\Rightarrow$ $H(j\omega)$ as fn of ω
polar coordinate
One plot: $|H(j\omega)|$ vs ω
$\angle H(j\omega)$ vs ω


PPS
V_{in}
V_o
$H(s) = \frac{1}{sCR+1}$
Goal: $H(j\omega) = \frac{1}{j\omega(RC)+1}$
time constant
$RC = \frac{1}{\omega_c}$
$= \frac{1}{j(\omega/\omega_c)+1}$
cut-off freq
$|H(j\omega)| = \frac{1}{\sqrt{1+\frac{\omega^2}{\omega_c^2}}}$
$\angle H(j\omega) = -\tan^{-1}\frac{\omega}{\omega_c}$
$\log_{10}|H(j\omega)|$
20
$-20\log\frac{\omega}{\omega_c}$
$\frac{\omega_c}{\omega}$
0
$\log\frac{\omega}{\omega_c}$
$\omega = \omega_c$
$\omega \gg \omega_c$
Bode
$\log|H(j\omega)| = -\frac{1}{2}\log_{10}\left(1+\frac{\omega^2}{\omega_c^2}\right)$
$\omega \ll \omega_c$ $\omega \sim 0$ $= 0$
$\omega \gg \omega_c$ $-\frac{1}{2}\cdot 2\cdot\log_{10}\frac{\omega}{\omega_c} = -\log_{10}\frac{\omega}{\omega_c}$ ← str. line
w/ diagonal slope
Generally plotted $20\log_{10}|H(j\omega)|$
$|0|_{10}$ ( ) dB
2x because $|H(j\omega)|$ is V/C gain
power gain is $|H(j\omega)|^2 \rightarrow |0|_{10}$ (power gain)

Ideal bode plot

$20 \log_{10} |H(j\omega)|$

Bode plot →

slope −20dB per 10× increase in ω

$-20 \log_{10} \frac{\omega}{\omega_c}$

$\omega = \omega_c$ $\log \frac{\omega}{\omega_c}$ →

−20 dB falls per decade

Correction at $\omega = \omega_c$

$20 \log_{10} |H(j\omega)| = 20 \log_{10} \frac{1}{\sqrt{1+1}}$

$\omega \to 10\omega$

$= -10 \log_{10} 2$

~ -3.01

$\frac{1}{\sqrt{2}} = 2^{-1/2}$

(Passive) Circuits ⇒ LPF

PS7

1st order

PS8

Frequency filter → allows one part of the spectrum blocks remainig

(LPF) Low pass filter → allows only low freq.

Ideally $20 \log_{10} |H(j\omega)|$

Ideally wanted (Can we get this from circuit)

Time domain response $h(t)$

Unbdd

non-Causal

sinc

Cutoff

$\times$ (Digitally possible)

Can we improve a RC LPF?

$s = j\omega$

2nd order -40 dB/dec $(=)$ $|H(j\omega)| \cdot \frac{1}{\sqrt{1+\left(\frac{\omega}{\omega_c}\right)^2}} \cdot \frac{1}{\sqrt{1+\left(\frac{\omega}{\omega_c}\right)^2}}$

3rd order $= -60$ dB/dec $(=)$

-20 dB/decade — 1st

slope $\downarrow 2x$ -40 dB/decade

$3x$ -60 dB/

$\omega_c = \frac{1}{RC}$

R 10R 100R

C

pss LTI $(=)$ $\frac{1}{(s/\omega_c + 1)^3}$

psw

R_1 R_2

C_1 C_2 $(=)$ $\frac{1}{sR_1C_1+1} \cdot \frac{1}{sR_2C_2+1}$ if $R_2 >> R_1$

$=$

lot of loss $\Rightarrow$ not C_1 active circuit (OPAMP)

Bode plot can be used for any transfer fn. $C_1/2$

$\left| \frac{1}{j\omega(R_1C_1)+1} \cdot \frac{1}{j\omega R_2C_2+1} \right| = \left| \frac{1}{j\,\omega/\omega_1 + 1} \cdot \frac{1}{j\,\omega/\omega_2 + 1} \right|$

ω_1 $1/\omega_2$

